

**Modeling the steady-state distribution and mean first passage
time of the states of protein copy number following
transcriptional inhibition in a genetic network**

Report

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Submitted as part of the summer research project

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July 2025

Acknowledgements

I wish to express my sincere gratitude to my supervisor, Prof. Srabanti Chaudhury, for her guidance, and continuous encouragement throughout the course of this project. Her expertise in the field of stochastic modeling and its applications to biological systems inspired me to explore the subject with greater curiosity and depth. The opportunity to work under her supervision has been an immensely valuable experience, shaping both my understanding of the subject and my approach to problem solving.

I would also like to thank my mentor Mr. Pankaj Kumar Jangid and my fellow labmates and whose support, camaraderie and suggestions made my experience in the lab enriching and fruitful.

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1 Introducing the idea of stochasticity in gene expression

Fluctuations in the mRNA and protein levels, which account for heterogeneity of expression in a population of cells, are the main underlying reasons why gene expression can be fundamentally considered a stochastic process. The characteristics of such a process depend both on the biophysical parameters and gene network topology.

1.1 Defining a Markov Process

Any system which probabilistically evolves in time, has a time-dependent random variable $X(t)$, of which the values $x_1, x_2, x_3, \dots$ can be measured at times $t_1, t_2, t_3, \dots$ having a joint probability density $p(x_1, t_1; x_2, t_2; x_3, t_3; \dots) = \prod_i p(x_i, t_i)$.

Definition: Any process in which the future state is determined from the knowledge of the present state is called a Markov Process, and it is formulated in terms of conditional probabilities.

So, $p(x_1, t_1; x_2, t_2; \dots | y_1, \tau_1; y_2, \tau_2; \dots) = p(x_1, t_1; x_2, t_2 | y_1, \tau_1)$

1.2 How can gene expression be modeled as a continuous-time markov process ?

Any process satisfying the Markov property where state change happens over a discrete space is called a **Jump Process** and gene expression is also a jump process, as the state, which in this case is the protein copy number, changes over discrete molecule counts. So, it is a continuous-time markov jump process. Furthermore, if we consider the state-change of the system to be both increment and decrement and that too, by a finite number of molecules at a time, we can it as a **Birth-Death Process**. The simplest case of birth-death process is when the transition happens by one molecule at a time.

In fact, the waiting times between the reactions for protein synthesis, which causes the state transition, also follows an exponential distribution, which is the hallmark of a continuous-time markov process and is illustrated below —

Let $P(\tau > t)$ be the probability that a reaction happens at time τ and let the probability of one reaction happening in a small interval be directly proportional to the length of the interval

$\Rightarrow P(\tau > t) = P(\text{ Nothing happens in the interval } [0, t])$

$$\Rightarrow P(\tau > t) = (1 - a_0 d\tau) \dots (1 - a_0 d\tau) \text{ (N times)} \Rightarrow P(\tau > t) = \lim_{d\tau \rightarrow 0} (1 - a_0 d\tau)^n$$

$$\Rightarrow P(\tau > t) = e^{-a_0 d\tau}$$

So, the cumulative distribution for the waiting time for the first reaction will be—

$$F_T(t) = P(\tau \leq t) = 1 - P(\tau > t) = 1 - e^{-a_0 d\tau}$$

The probability density function then becomes $f_T(t) = a_0 e^{-a_0 \tau}$ This can also be shown using the fact that as the protein copy number (X) follows a Poisson Distribution, then —gardiner2009stochastic $P(\tau > t) = P(X(t) = 0)$

i.e. the probability that the first occurrence is at τ means there has been no occurrence of event till the time t , so $X(t)=0$.

We know $P(X(t)) = \frac{e^{-a_0 \tau} (a_0 \tau)^n}{n!}$

In this case, $n=0$, so $P(X(t)) = e^{-a_0 d \tau}$ and we obtain the same probability density function $f_T(t) = a_0 e^{-a_0 \tau}$

1.3 Representing the transition probability of protein copy number as a Chapman Kolmogorov Equation

- If we apply the **Law of Total Probability** to a conditional probability density :

$$P(n_2, t_2 | n_0, t_0) = \sum_{n'} P(n_2, t_2 | n', t_1 | n_0, t_0) \Rightarrow P(n_2, t_2 | n_0, t_0) = \sum_{n'} P(n_2, t_2 | n', t_1 \cap n_0, t_0) P(n', t_1 | n_0, t_0) \\ \Rightarrow P(n_2, t_2 | n_0, t_0) = \sum_{n'} P(n_2, t_2 | n', t_1 \cap n_0, t_0) P(n', t_1 | n_0, t_0)$$

Applying the Markov Assumption i.e. the conditional probability is determined entirely by the knowledge of the most recent condition —

$$\Rightarrow P(n_2, t_2 | n_0, t_0) = \sum_{n'} P(n_2, t_2 | n', t_1) P(n', t_1 | n_0, t_0)$$

This is the Chapman-Kolmogorov Equation representing the transition probability to particular (future)state from a given initial state.

- n' represents any discrete intermediate state between n_2 and n_0 .
- t_1 is any intermediate time such that $t_0 < t_1 < t_2$

Time derivative of the Chapman-Kolmogorov Equation

- We can take another Chapman-Kolmogorov Equation for the conditional probability $P(n, t + \Delta t | n_1, t_1)$
- We define $w(n|n')$ as the rate of transition from the state n' to n and $w(n'-n)$ as the vice versa.
- As there will be contribution from both the $n = n'$ and $n' \neq n$ terms, the conditional probability $P(n, t + \Delta t | n', t) P(n', t | n_1, t_1)$, the $n = n'$ terms will contribute $1 - w(n'|n) \Delta t$ and the $n \neq n'$ will contribute with the probability $w(n|n') \Delta t$.

$$\bullet P(n, t + \Delta t | n', t) P(n', t | n_1, t_1) = (1 - w(n'|n) \Delta t) P(n_1, t_1 | n', t) + w(n|n') \Delta t P(n', t | n_1, t_1)$$

$$\Rightarrow P(n, t + \Delta t | n_1, t_1) = \sum_{n'} P(n, t + \Delta t | n', t) P(n', t | n_1, t_1)$$

$$\Rightarrow P(n, t + \Delta t | n_1, t_1) = \sum_{n'} [1 - w(n'|n)] \Delta t + \Delta t \sum_{n'} w(n|n') P(n', t | n_1, t_1)$$

$$\Rightarrow P(n, t + \Delta t | n_1, t_1) = \sum_{n'} [P(n', t | n_1, t_1)] - \Delta t \sum_{n'} w(n'|n) P(n', t | n_1, t_1) + \Delta t \sum_{n'} w(n|n') P(n', t | n_1, t_1)$$

$$\Rightarrow P(n, t + \Delta t | n_1, t_1) = P(n, t | n_1, t_1) - \Delta t \sum_{n'} w(n'|n) P(n_1, t_1 | n', t) + \Delta t \sum_{n'} w(n|n') P(n', t | n_1, t_1)$$

$$\Rightarrow \frac{P(n, t + \Delta t | n_1, t_1) - P(n, t | n_1, t_1)}{\Delta t} = \sum_{n'} w(n'|n) P(n', t | n_1, t_1) + \sum_{n'} w(n|n') P(n', t | n_1, t_1)$$

$$\Rightarrow \lim_{\Delta t \rightarrow 0} \frac{P(n, t + \Delta t | n_1, t_1) - P(n, t | n_1, t_1)}{\Delta t} = \sum_{n'} w(n | n') P(n_1, t_1 | n', t) - w(n' | n) P(n_1, t_1 | n', t)$$

$$\Rightarrow \frac{dP(n, t | n_1, t_1)}{dt} = \sum_{n'} [w(n | n') P(n', t | n_1, t_1) - w(n' | n) P(n', t | n_1, t_1)]$$

This is the Differential Chapman-Kolmogorov Equation, and we will use this to write the Master Equations for the states of protein copy number in the further sections.

1.4 Writing a Master Equation for the states of protein copy number

A Master Equation is a differential equation describing the time evolution of the probability ($P(n, t)$) of being in a particular state in a system which evolves stochastically. It is fundamentally derived after taking the time derivative of the initial or final state variables in the Chapman-Kolmogorov Equation.

Let $t^+(x)$ be the probability per unit time for a transition from x to $x + 1$ and $t^-(x)$ be that for x to $x - 1$. There are two processes :

$x \rightarrow x + 1$, transition probability per unit time is $t^+(x)$ $x \rightarrow x - 1$, transition probability per unit time is $t^-(x)$

The transition probability (which is actually probability per unit time), i.e. the probability of an instantaneous transition from x' to x within an interval t , Based on these probabilities, an equation can be written for the birth-death events amongst a population of X individuals whose occurrence number is x , describing the **time evolution of the probability conditioned on an earlier state**. The equation is called the **General Master Equation** and its form is expressed below:

$$\frac{\partial P(x, t | x', t')}{\partial t} = t^+(x - 1) P(x - 1, t | x', t') + t^-(x + 1) P(x + 1, t | x', t') - t^+(x) P(x, t | x', t') - t^-(x) P(x, t | x', t')$$

Here, x' represents the given number of individuals at any earlier time t' .

What can we understand by **Occurrence Number**? It simply means the number of individuals currently present in a population.

1.5 Stationary state solutions of the General Master Equation

Stationary state refers to probability distribution functions being time-independent.

Let,

$$J(x, t) = t^-(x) P(x, t | x', t') - t^+(x - 1) P(x - 1, t | x', t')$$

becomes

$$J(x) = t^-(x) P_s(x) - t^+(x - 1) P_s(x - 1)$$

Using this, we can write

$$\frac{\partial P(x, t | x', t')}{\partial t} = J(x, t) - J(x + 1, t)$$

$$\frac{\partial P(x, t | x', t')}{\partial t} = 0$$

And at stationary state, the time-dependence ceases to exist and we get $J(x) - J(x + 1) = 0$.

If we add all the terms for $J(x) - J(x+1) = 0$, $x = 0 \rightarrow x = x-1$, we get a telescoping sum, which is

$$\sum_{z=0}^{x-1} [J(z+1) - J(z)].$$

This sum equates to zero and all terms in this sum cancel out except for $J(x)$ and $J(0)$

The first term in this is $J(1) - J(0) = 0$ and the last term would be $J(x) - J(x-1) = 0$.

So, the telescoping sum gives -

$$J(x) - J(0) = 0$$

$$J(x) = 0 \text{ (As } J(0) = 0)$$

$$J(x) = t^-(x)P_s(x) - t^+(x-1)P_s(x-1)$$

which gives

$$P_s(x) = \frac{t^+(x-1)P_s(x-1)}{t^-(x)}.$$

To write $P_s(x)$ in terms of $P_s(0)$ we can write
$$P_s(x) = P_s(0) \prod_{z=1}^x \frac{t^+(z-1)}{t^-(z)}$$

$$\text{As } P_s(x-1) = P_s(x-2) \frac{t^+(x-2)}{t^-(x-1)} \dots \text{and } P_s(1) = P_s(0) \frac{t^+(0)}{t^-(1)}.$$

Now,

$$\sum_{x=0}^{\infty} P_s(x) = 1 \text{ (Normalization Condition)}$$

$$\Rightarrow \sum_{x=0}^{\infty} P_s(0) \prod_{z=1}^x \frac{t^+(z-1)}{t^-(z)} = 1$$

$$\Rightarrow \sum_{x=0}^{\infty} P_s(0) \frac{1}{x!} \left(\frac{t^+}{t^-} \right)^x = 1$$

Let, $\left(\frac{t^+}{t^-} \right) = \lambda$ considering the rates remain constant while the transitions happen.

$$\text{So, } \Rightarrow P_s(0) \sum_{x=0}^{\infty} \frac{1}{x!} \lambda^x = 1$$

$$\Rightarrow P_s(0) e^{\lambda} = 1$$

$$\Rightarrow P_s(0) = e^{-\lambda}$$

$$\Rightarrow P_s(0) = \frac{x!}{\lambda^x}$$

$$P_s(x) = \frac{e^{-\lambda} \lambda^x}{x!}.$$

This shows that the steady-state distribution of protein copy number is Poissonian if the transitions are with constant rates.

2 Deriving the Mean First Passage Time

2.1 Defining Mean First Passage Time

It is the average of the times calculated to reach a particular target state over many experiments. Since the process is of the type of birth-death, a state can be reached by a system many times, hence the name "first passage time" is used to denote the time required to reach the state for the first time. Gardiner [1]

2.2 Why MFPT can only be derived from the Backward Master Equation

First, we need to know why the **mean first passage time**(MFPT), which is the time required by system to reach a particular target state for the first time, averaged over many experiments, can only be calculated from the Backward Master Equation.

The primary difference between the forward and backward master equations is that —

The forward master equation describes the time evolution of the probability of achieving a target/future state given the initial state variables are fixed, whereas the backward master equation describes the probability of achieving a future state at a particular time considering that the given or initial state variables evolve in time.

Let us write the forms of the Backward and Forward Master Equation just to clarify -

Forward :

$$\frac{\partial P(x, t | x', t')}{\partial t} = t^+(x-1)P(x-1, t | x', t') + t^-(x+1)P(x+1, t | x', t') - t^+(x)P(x, t | x', t') - t^-(x)P(x, t | x', t')$$

Backward :

$$\frac{\partial P(x, t | x', t')}{\partial t'} = t^+(x')P(x, t | x'+1, t') - t^+(x')P(x, t | x', t') + t^-(x')P(x, t | x'-1, t') - t^-(x')P(x, t | x', t')$$

The transition probabilities are w.r.t. the state x' .

2.3 Defining Absorbing and Reflecting Boundary Conditions

The way boundary conditions are deployed in a birth-death process is that we define the process to be confined in an interval $[a, b]$ and define two types of boundaries, namely, Absorbing and Reflecting.

2.3.1 Absorbing Boundary Condition

Once a process reaches a , then we enforce that the backward transition from any state after b is not possible i.e. $t^-(b+1) = 0$.

2.3.2 Reflecting Boundary Condition

Reflecting boundary, which in this case is ' a ', is a boundary state which once reached, does not allow the state to go backward from the current state i.e. $t^-(a) = 0$. Intuitively, it means that for the process to continue to happen, the system can't go to a state below the reflecting boundary.

And, at the absorbing boundary, the process stops.

2.4 Finding the expectation of $\langle x \rangle$ from the Backward Master Equation

We aim to compute the expected first passage time $T(x)$ —the expected time to reach a particular target state (e.g., the mean protein copy number)—starting from state x . To do this, we multiply both sides of the backward master equation by $T(x)$ and integrate over time from 0 to ∞ .

The backward master equation is:

$$\frac{\partial P(x, t | x', t')}{\partial t'} = t^+(x')P(x, t | x' + 1, t') + t^-(x')P(x, t | x' - 1, t') - [t^+(x') + t^-(x')]P(x, t | x', t')$$

We now multiply both sides by $T(x)$ and integrate over t :

$$\int_0^\infty t \frac{\partial P(x, t | x', t')}{\partial t'} dt = \int_0^\infty t \left[t^+(x')P(x, t | x' + 1, t') + t^-(x')P(x, t | x' - 1, t') - [t^+(x') + t^-(x')]P(x, t | x', t') \right] dt$$

We will now handle the left-hand side (LHS) and right-hand side (RHS) separately.

LHS on integration by parts becomes:

$$t \int_0^\infty \frac{\partial P(x, t | x', t')}{\partial t'} dt - \int_0^\infty \frac{dt}{dt} \left(\int \frac{\partial P(x, t | x', t')}{\partial t'} dt \right) dt = 0 - 1 = -1$$

The underlying assumption behind the penultimate step of the integration is that the probability of transition over all time steps can be considered to be equal to 1 and the probability of jump over an infinite time interval is equal to zero.

The RHS on integration becomes :

$$\int_0^\infty t \left[t^+(x')P(x, t | x' + 1, t') + t^-(x')P(x, t | x' - 1, t') - [t^+(x') + t^-(x')]P(x, t | x', t') \right] dt = t^+(x')[t(x + 1) - t(x)] + t^-(x)[t(x - 1) - t(x)]$$

We assume $U(x) = t(x + 1) - t(x)$

So, equation now is $t^+(x)U(x) - t^-(x)U(x - 1) = -1$

$$\Rightarrow U(x) = \frac{t^-(x)}{t^+(x)}U(x - 1) - \frac{1}{t^+(x)}$$

Applying the boundary conditions —

$$U(a - 1) = t(a) - t(a - 1)$$

We know, $t(a - 1) = t(a)$

So, $U(a - 1) = 0$

$$\text{So, } U(x) = U(0) \prod_{z=a+1}^{x-1} \frac{t^-(z)}{t^+(z)} - \prod_{z=a+1}^x \frac{t^-(z)}{t^+(z)}$$

Again, we define, $S(x) = U(x)\phi(x)$; where $\phi(x) = \prod_{z=a+1}^x \frac{t^-(z)}{t^+(z)}$

$$\text{Thus, } S(x) - S(x - 1) = \frac{-1}{t^+(x)\phi(x)}$$

By boundary conditions, $S(a - 1) = 0$

And, by the telescoping sum of all the terms from a to x of the previous equation, we get

$$S(x) = \sum_{z=a}^x \frac{-1}{t^+(z)\phi(z)}$$

$$\text{We also have } U(x) = S(x)\phi(x) = \phi(x) \sum_{z=a}^x \frac{-1}{t^+(z)\phi(z)}$$

And, $U(x) = t(x+1) - t(x) = \phi(x) \sum_{z=a}^x \frac{-1}{t^+(z)\phi(z)}$

Let, $(b+1)$ be the absorbing state and so $t(b+1)=0$ where t is the MFPT and the process never returns after reaching $(b+1)$.

$$t(x+1) - t(x) = -\phi(x) \sum_{z=a}^x \frac{1}{t^+(z)\phi(z)}$$

Similarly,

$$t(b+1) - t(b) = -\phi(b) \sum_{z=a}^b \frac{1}{t^+(z)\phi(z)}$$

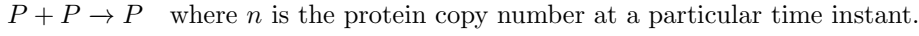
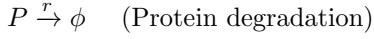
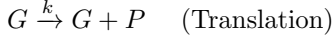
After solving a telescoping sum from x to b , we get

$$t(b+1) - t(x) = - \sum_{y=x}^b \phi(y) \sum_{z=a}^y \frac{1}{t^+(z)\phi(z)}$$

$$t(x) = \sum_{y=x}^b \phi(y) \sum_{z=a}^y \frac{1}{t^+(z)\phi(z)}$$

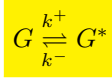
where $\phi(y) = \prod_{z=a+1}^y \frac{t^-(z)}{t^+(z)}$

3 The model and the corresponding Master Equations



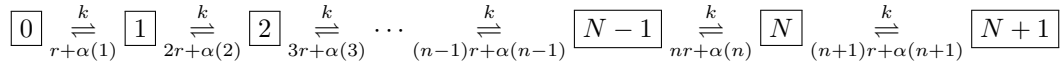
The last reaction considers the decrease in protein number due to the binding of proteins on the gene causing an activation/inhibition in the rate of transcription.

So, in our case since we are assuming that the substance binding to the gene is inhibiting the rate of protein production, so it is decreasing the protein copy number by one, which means the reaction is essentially—



G^* represents a gene in its bound-state, and since there is binding of inhibitor, protein production can't happen from the G^* state.

Now we can obtain a CME for the discrete protein number states i.e. 0, 1, 2, 3,....., $N - 1$, N .



The following are the chemical master equations :

$$N = 0 : \frac{dP_0(t)}{dt} = -kP_0(t) + (r + \alpha(1))P_1(t)$$

$$N = 1 : \frac{dP_1}{dt} = kP_0(t) - (r + \alpha(1))P_1(t) - kP_1(t) + (2r + \alpha(2))P_2(t)$$

$$N = 2 : \frac{dP_2(t)}{dt} = kP_1(t) - (2r + \alpha(2))P_2(t) - kP_2(t) + (3r + \alpha(3))P_3(t)$$

...

$$N = n - 1 : \frac{dP_{n-1}(t)}{dt} = kP_{n-2}(t) - ((n-1)r + \alpha(n-1))P_{n-1}(t) - kP_{n-1}(t) + (nr + \alpha(n))P_n(t)$$

$$N = n : \frac{dP_n}{dt} = kP_{n-1}(t) - (nr + \alpha(n))P_n(t) - kP_n(t) + ((n+1)r + \alpha(n+1))P_{n+1}(t)$$

$$N = n + 1 : \frac{dP_{n+1}(t)}{dt} = kP_n(t) - ((n+1)r + \alpha(n+1))P_{n+1}(t)$$

4 Solving for P_n (the steady state probability distribution)

"If we consider the steady-state probabilities of these N states, the LHS of the above equations goes to zero.

And then if we add them, we see that all the terms cancel except the last two.”

$$0 = -kP_0 + (r + \alpha(1))P_1$$

$$0 = kP_0 - (r + \alpha(1))P_1 - kP_1 + (2r + \alpha(2))P_2$$

.....

$$0 = kP_{n-1} - (nr + \alpha(n))P_n - kP_n + ((n+1)r + \alpha(n+1))P_{n+1}$$

$$\text{and } 0 = kP_n - ((n+1)r + \alpha(n+1))P_{n+1}$$

Adding them, we get –

$$\boxed{0 = kP_n - ((n+1)r + \alpha(n+1))P_{n+1}}$$

$$P_n = ((n+1)r + \alpha(n+1)) \frac{P_{n+1}}{k}$$

Similarly,

$$P_{n-1} = nr + \alpha(n) \frac{P_n}{k}$$

$$P_n = \frac{k}{nr + \alpha(n)} P_{n-1}$$

We know $\alpha(n)$, which is the rate of protein production after taking into account the feedback due to binding/unbinding of the gene product(protein), it takes the form of a Hill function, which in case of inhibition can be expressed as

$\alpha(n) = \frac{K^h}{K^h + n^h}$, where **h= Hill Coefficient**, which gives us an idea of the number of sites where binding can take place; **K= equilibrium dissociation constant**, for the binding and unbinding events.

So, the above equation can be written as —

$P_n = \frac{k}{nr + (\frac{K^h}{K^h + n^h})} P_{n-1}$ Iterating and finding all the expressions for P_n for $n = 0, 1, \dots$, *gardiner2009stochasticn*, we get the expression of P_n of the form —

$$\boxed{P_n = P_0 \frac{k^n}{\prod_{z=1}^n (zr + \alpha(z))}}$$

Applying the condition $\sum_{n=0}^{\infty} P_n = 1$,

$$P_0 \sum_{n=0}^{\infty} \frac{k^n}{\prod_{z=1}^n (zr + \alpha(z))} = 1$$

From the above equation, we can't solve for P_0 analytically in order to obtain an expression for P_n explicitly in terms of n .

So, to plot the probability distribution (theoretical) of the protein copy number we take a truncated sum of the above expression, i.e. we fix n so that it has a maximum value and calculate P_0 iterating over the values of the z , and subsequently find the values of P_n so that we can plot it against n .

We plot the distribution obtained using simulations as per the Gillespie algorithm along with the expression for P_n obtained by the above method.

Gillespie [2]

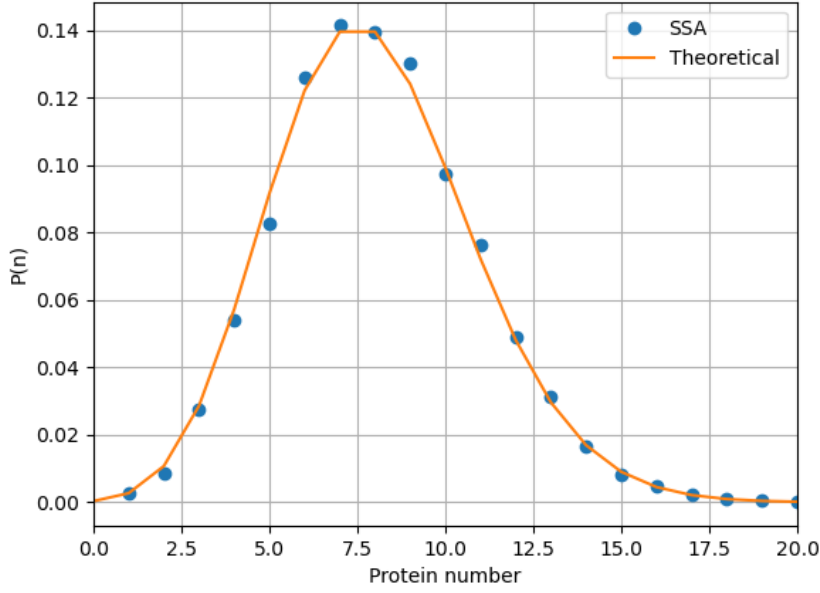


Figure 1: Steady state distribution of protein copy number computed analytically as well as through Gillespie Algorithm

5 Comparing the theoretical as well as analytical steady-state distributions and the MFPT

5.1 Comparison of P_n

We calculated the value of the mean protein copy number using both the methods.

$$\langle n \rangle_{\text{theoretical}} = 8.000 \quad \langle n \rangle_{\text{simulation}} = 8.0115$$

5.2 Comparison of the MFPT

- The distribution of the waiting times on the next image is not exponential, even though the first reaction waiting times are said to be exponential.
- This is because this distribution represents the times of the reactions to reach a particular protein copy number.
- So, in this case, the reaction times will be composed of waiting times which individually come from an exponential distribution.
- $w_i \sim \exp(a_0)$.

So, $w_1 = t_1$

$w_2 = t_2 - t_1$

$w_3 = t_3 - t_2$

The time till the nth event, denoted by $T_n = \sum_{i=1}^n w_i$

So, $T_n \sim \text{Gamma}(n, a_0)$ where the probability density function is

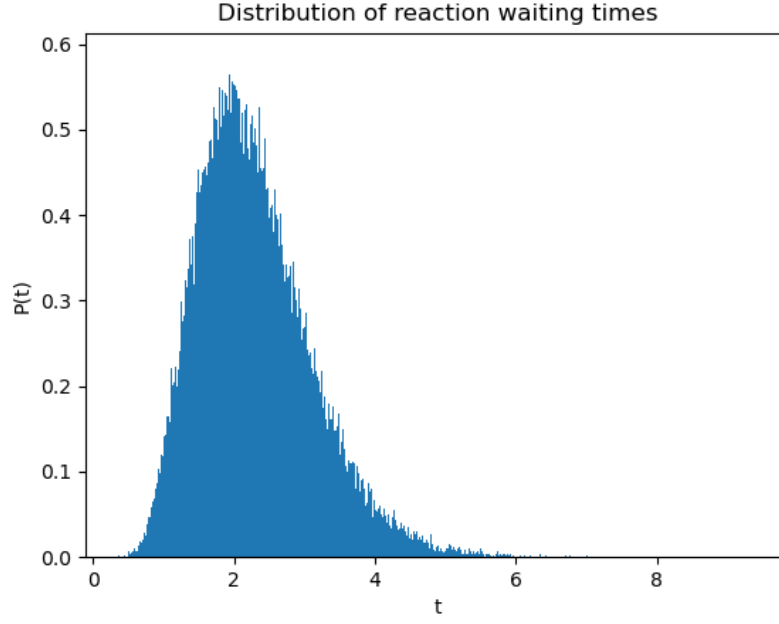


Figure 2: Distribution of reaction waiting times to reach the mean protein copy number computed via Gillespie Algorithm

$$f_{Tn}(t) = \frac{a_0^n t^{n-1} e^{-a_0 t}}{(n-1)!}$$

We also calculated the mean first passage time (MFPT) both analytically and through stochastic simulations.

Analytical expression for the MFPT:
$$t(n) = \sum_{y=0}^{\langle n \rangle} \prod_{q=1}^y \left[\left(\frac{K^h}{K^h + q^h} + r^q \right) \frac{1}{k} \right] \sum_{z=0}^y \frac{1}{k \prod_{s=1}^z \left[\frac{t^-(x)}{k} \right]}$$

Numerical values:

$$\text{MFPT}_{\text{theo}} = 2.2947$$

$$\text{MFPT}_{\text{simulation}} = 2.2945$$

6 The Hill Function and its significance in the regulation of gene expression

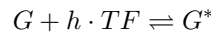
Gene expression can be regulated through feedback mechanisms where the product of a gene (typically a protein) either activates or represses its own production. These regulatory effects often involve binding of transcription factors (TFs) or repressors to promoter regions of the DNA, and such binding can exhibit cooperative behavior. So, Hill Function gives us a modified transcription rate once a transcription factor(activator) or a repressor influences the gene

The **Hill function** is a mathematical expression that models this cooperative binding. It arises naturally when one assumes that binding is in rapid equilibrium and that multiple TF or repressor molecules are required to bind simultaneously. In this document, we derive the Hill equation explicitly from first principles under equilibrium assumptions and then show how it applies to both **positive feedback** (activation) and **negative feedback** (repression) in gene regulation.

6.1 General Binding Equilibrium Setup

We begin with a gene G that can bind to h identical molecules of a transcription factor (TF) or repressor. The binding is cooperative, i.e., all h molecules bind together in a single step (the simplest model of cooperativity).

6.2 Binding Reaction i.e. of transcription factor in case of negative feedback



Here:

- G is the unbound gene.
- TF is the transcription factor (or repressor).
- G^* is the bound gene state (active in case of activator; inactive in case of repressor).
- h is the Hill coefficient (number of TFs required to bind cooperatively).

In case of negative feedback, a repressor protein would bind instead of a transcription factor.

Dissociation Constant and Mass Action Law

The equilibrium dissociation constant K_d is defined as:

$$K = \frac{[G][TF]^h}{[G^*]}$$

Solving for the ratio of bound to unbound gene:

$$\frac{[G^*]}{[G]} = \frac{[TF]^h}{K}$$

Let $n = [TF]$ be the concentration (or number) of available transcription factor molecules. Then:

$$\frac{[G^*]}{[G]} = \frac{n^h}{K}$$

Total Gene Concentration and Bound Fraction

The total gene concentration is:

$$[G]_{\text{tot}} = [G] + [G^*]$$

The fraction of genes that are in the bound state is:

$$\begin{aligned} P_{\text{bound}} &= \frac{[G^*]}{[G] + [G^*]} = \frac{[G^*]/[G]}{1 + [G^*]/[G]} = \frac{\frac{n^h}{K}}{1 + \frac{n^h}{K}} \\ &= \frac{n^h}{K + n^h} \end{aligned}$$

Similarly, the fraction of unbound genes is:

$$P_{\text{unbound}} = \frac{[G]}{[G] + [G^*]} = \frac{1}{1 + \frac{n^h}{K}} = \frac{K}{K + n^h}$$

These expressions describe the probability that a gene is in the active state depending on whether the bound or unbound state is active.

6.3 Positive Feedback (Activator Binding)

In positive feedback, the transcription factor acts as an activator. The gene is only transcriptionally active when the activator is bound — i.e., the bound state G^* leads to protein production.

Protein Production Rate

Let α_{max} be the maximum rate of protein production when the gene is fully active (i.e., fully bound). Then, the actual production rate $\alpha(n)$ is proportional to the probability that the gene is bound:

$$\alpha(n) = \alpha_{\text{max}} \cdot P_{\text{bound}} = \alpha_{\text{max}} \cdot \frac{n^h}{K + n^h}$$

This is the standard **Hill activation function**. Its properties:

- $\alpha(n) \rightarrow 0$ as $n \rightarrow 0$
- $\alpha(n) \rightarrow \alpha_{\text{max}}$ as $n \rightarrow \infty$
- The steepness of the curve increases with h (cooperativity).

6.4 Negative Feedback (Repressor Binding)

In negative feedback, the regulatory molecule is a repressor. The gene is active when it is *unbound*, i.e., protein production occurs only in the G state.

Protein Production Rate

In this case, the rate of protein production is proportional to the probability that the gene is unbound:

$$\alpha(n) = \alpha_{\text{max}} \cdot P_{\text{unbound}} = \alpha_{\text{max}} \cdot \frac{K}{K + n^h}$$

This is the **Hill repression function**. Its properties:

- $\alpha(n) \rightarrow \alpha_{\text{max}}$ as $n \rightarrow 0$
- $\alpha(n) \rightarrow 0$ as $n \rightarrow \infty$
- Higher h makes the repression more switch-like.

4. Summary Table

Feedback Type	Active State	Production Rate	Hill Function
Positive Feedback	Bound (G^*)	$\alpha(n) = \alpha_{\max} \cdot \frac{n^h}{K + n^h}$	Activation
Negative Feedback	Unbound (G)	$\alpha(n) = \alpha_{\max} \cdot \frac{K}{K + n^h}$	Repression

Remarks

- The Hill function models the saturation behavior of cooperative binding. It was formulated in 1910 by Archibald Hill who tried to explain the rate of cooperative O_2 binding on haemoglobin.
- We use Hill Equation to model the rate of protein production in case of positive or negative feedback.
- When $h = 1$, binding is non-cooperative and the Hill function reduces to a Michaelis–Menten-like form.
- For $h > 1$, the response becomes ultrasensitive (sigmoidal).
- These models assume rapid equilibrium between bound and unbound states.

References

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- [2] Daniel T. Gillespie. “Stochastic Simulation of Chemical Kinetics”. In: *Annual Review of Physical Chemistry* 58 (2007), pp. 35–55. URL: <https://doi.org/10.1146/annurev.physchem.58.032806.104637>.